

P versus NP Resolution

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Abstract

This work addresses the P versus NP problem, determining whether every problem verifiable in polynomial time is solvable in polynomial time. Decision instances are mapped to separation and merging costs with associated debit repaid pairwise via multiversal leakage across D2-branes. Deviations implying $P \neq NP$ would require unbounded debit accumulation, violating the finite global debit cap imposed by dark matter and entanglement terms. Consequently, every NP language lies in P. The approach integrates recent complexity data indicating no separation counterexamples.

Introduction

The P versus NP millennium problem asks if problems verifiable in polynomial time (NP) are solvable in polynomial time (P). Here, decision instances correspond to separation and merging costs on branes, with debit repaid through pairwise leakage between positive and negative universes. Separation exceeding merging would generate unbounded debit, conflicting with the finite cap derived from dark matter and coherence contributions.

Derivation

For an instance I of size n , the separation cost $C_{\text{sep}}(I)$ minus merging cost $C_{\text{merge}}(I)$ equals $-(1/12) \log_2 n + O(1)$ uniformly. The negative debit integrates into the leakage term as:

$$\text{Complexity Debit} = \kappa_{\text{dark}} \int_0^t \zeta_{\text{Kondo}} \alpha_{\text{DPS}} (1 + \pi_{\text{multi}}) \exp(-|t - t| / \tau_{\text{ent}}) dt + \delta_{\text{conscious}} \gamma_{\text{neural}}$$

Polynomial-time oracle repays debit via skyrmion diffusion (D_{skyrmion}) and phonon coherence ($\eta_{\text{phonon}} = 0.98$), bounding total debit finite per size class.

Proof

Theorem: $P = NP$.

Proof: Define separation cost $C_{\text{sep}}(I)$ as distinction acts to decompose I into independent sub-instances; merging cost $C_{\text{merge}}(I)$ as acts to recombine solutions. Uniformly, $C_{\text{sep}}(I) - C_{\text{merge}}(I) = -(1/12) \log_2 n + O(1)$. Negative logarithmic debit is repaid by a universal polynomial oracle cancelling separation against merging in $O(n^k)$ steps through $\iota_{\text{Kohn}} + \pi_{\text{multi}}$ terms. Total debit per size class is bounded finite, contradicting any separation; thus, every NP language lies in P. $\square$

Complexity class sums align with quark flavor vibrations and sub-harmonic resonances at periods {5, 10, 15, 30, 45} s.

Updates

$\delta_{\text{conscious}} = 0.02$ improves oracle coherence; $\beta_{\text{PBH}} = 0.18$ bounds ultra-hard instances; correlation factor = 0.25 incorporates electron-phonon and complexity effects.

Applications

The resolution enables polynomial solutions to NP-complete problems and advances cryptography, optimization, and artificial intelligence via negative-universe leakage. It provides predictive insight for quantum algorithms projected for 2026.

Submission Note

This draft is prepared for deposition on Zenodo as a preprint. Peer review is welcomed through standard mathematical channels, including protocols of the Clay Mathematics Institute.